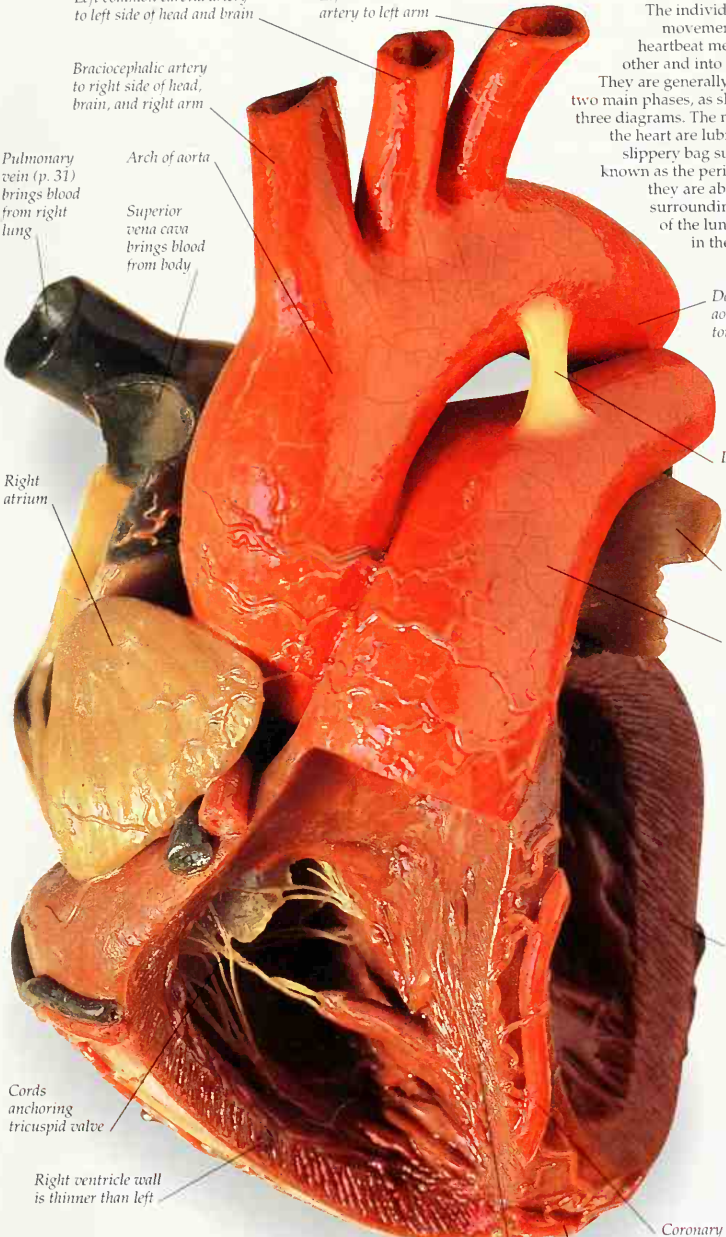
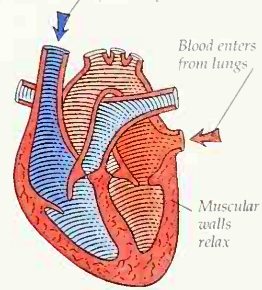
HOW THE HEART BEATS

The individual muscular movements of a single heartbeat merge into each other and into the next beat. They are generally divided into two main phases, as shown here by three diagrams. The movements of the heart are lubricated by the slippery bag surrounding it, known as the pericardium, and they are absorbed by the surrounding sponginess of the lungs (pp. 26-27) in the chest cavity.



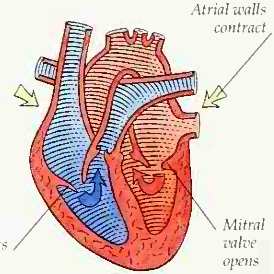
INSIDE THE HEART

This model shows the thickness of the muscular ventricle walls and the septum separating the right and left pumps. The atria are little more than flaplike pouches sitting on top of each ventricle, receiving blood at almost no pressure from the main veins. The coronary arteries supply blood to the heart muscle itself. Each time a ventricle contracts, as part of a heartbeat, it ejects about 0.12 pt (70 ml) of blood. In a day, the ventricles pump 2,200-3,300 gal (10,000-15,000 liters) of blood.



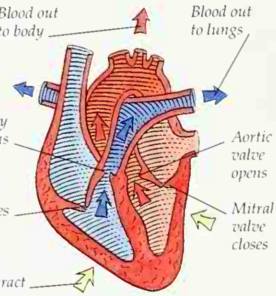
1 RELAX AND REFILL

The muscles in the walls of the heart relax. Blood seeps in under low pressure from the main veins, as it returns from the whole body (systemic circulation) and the lungs (pulmonary circulation). This relaxed, refilling phase is known as diastole.



2 ATRIAL CONTRACTION

The thin atrial walls squeeze blood through the valves into the ventricles. The wave of contraction is initiated by a clump of nerve-like cells in the right atrium wall, known as the sinoatrial node or natural pacemaker.



3 VENTRICULAR CONTRACTION

The wave of contraction jumps to the lower point, or apex, of the heart and spreads up through the ventricle muscle. Blood is forced under pressure up into the arteries. This high-pressure ejection phase is called systole. As the valves snap closed, they make the lub-dup sound of a heartbeat.